

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

Roll No.: A24126552260

Student Name: K. Sirisha

1. TITLE

JS Basics - Arrays & Functions

2. AIM

To understand JavaScript arrays and functions and use built-in array methods to process student marks.

3. OBJECTIVES

- 1. To create and initialize JavaScript arrays.**
- 2. To define and call user-defined functions.**
- 3. To calculate total and average using array processing.**
- 4. To filter values based on a condition.**
- 5. To transform array values using the map() method.**

4. THEORY

JavaScript arrays store multiple values in a single variable. Functions group reusable logic into blocks. Array methods such as reduce(), filter() and map() make it possible to calculate totals, select matching values and transform each element efficiently.

5. ALGORITHM/PROCEDURE

- 1. Create an array containing student marks.**
- 2. Define a function to calculate the total using reduce().**
- 3. Define a function to calculate the average.**
- 4. Use filter() to select marks greater than or equal to the pass mark.**
- 5. Use map() to convert marks into grades.**
- 6. Display the processed values in the webpage.**
- 7. Open the page in a browser and verify the output.**

6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Array & Function Demo</title>
</head>
<body>
<h1>Student Marks Analyzer</h1>
<p id="output"></p>
<script>
const marks = [78, 92, 65, 88, 74];

function total(values){
  return values.reduce((sum, mark) => sum + mark, 0);
}

function average(values){
  return total(values) / values.length;
}

function passed(values){
  return values.filter(mark => mark >= 70);
}

function grade(mark){
  if(mark >= 90) return "A+";
  if(mark >= 80) return "A";
  if(mark >= 70) return "B";
  return "C";
}

const grades = marks.map(grade);

document.getElementById("output").innerHTML =
  "Marks: " + marks.join(", ") + "<br>" +
  "Total: " + total(marks) + "<br>" +
  "Average: " + average(marks).toFixed(2) + "<br>" +
  "Passed: " + passed(marks).join(", ") + "<br>" +
  "Grades: " + grades.join(", ");
</script>
</body>
</html>
```

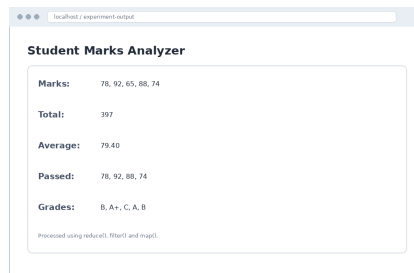
7. CODE EXPLANATION

Component	Purpose
const marks	Creates an array containing the student marks.
function total()	Calculates the sum of all array elements.
reduce()	Combines all array values into a single total.
average()	Divides the total by the number of elements.
filter()	Returns only marks satisfying the pass condition.
grade()	Converts a numerical mark into a grade.
map()	Creates a new array by applying grade() to each mark.
innerHTML	Displays the calculated results in the webpage.

8. TEST CASES

Test Case	Test Case 1
Input	marks = [78, 92, 65, 88, 74]
Expected Total	397
Expected Average	79.40
Expected Passed Marks	78, 92, 88, 74
Expected Grades	B, A+, C, A, B

Test Case 1 - Output



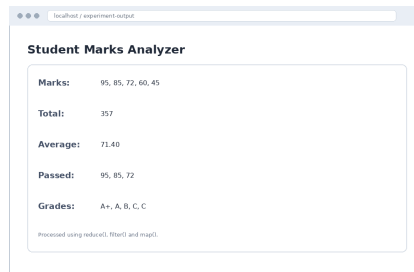
The screenshot shows a web browser window with the title 'localhost / experiment-output'. The page content is titled 'Student Marks Analyzer'. It displays the following information:

Marks:	78, 92, 65, 88, 74
Total:	397
Average:	79.40
Passed:	78, 92, 88, 74
Grades:	B, A+, C, A, B

At the bottom, it states: 'Processed using reduce(), filter() and map()'.

Test Case	Test Case 2
Input	marks = [95, 85, 72, 60, 45]
Expected Total	357
Expected Average	71.40
Expected Passed Marks	95, 85, 72
Expected Grades	A+, A, B, C, C

Test Case 2 - Output



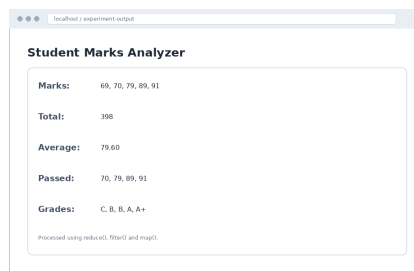
The screenshot shows a web browser window with the title 'localhost / experiment-output'. The page content is titled 'Student Marks Analyzer'. It displays the following information:

Marks:	95, 85, 72, 60, 45
Total:	357
Average:	71.40
Passed:	95, 85, 72
Grades:	A+, A, B, C, C

At the bottom, it states: 'Processed using reduce(), filter() and map()'.

Test Case	Test Case 3
Input	marks = [69, 70, 79, 89, 91]
Expected Total	398
Expected Average	79.60
Expected Passed Marks	70, 79, 89, 91
Expected Grades	C, B, B, A, A+

Test Case 3 - Output



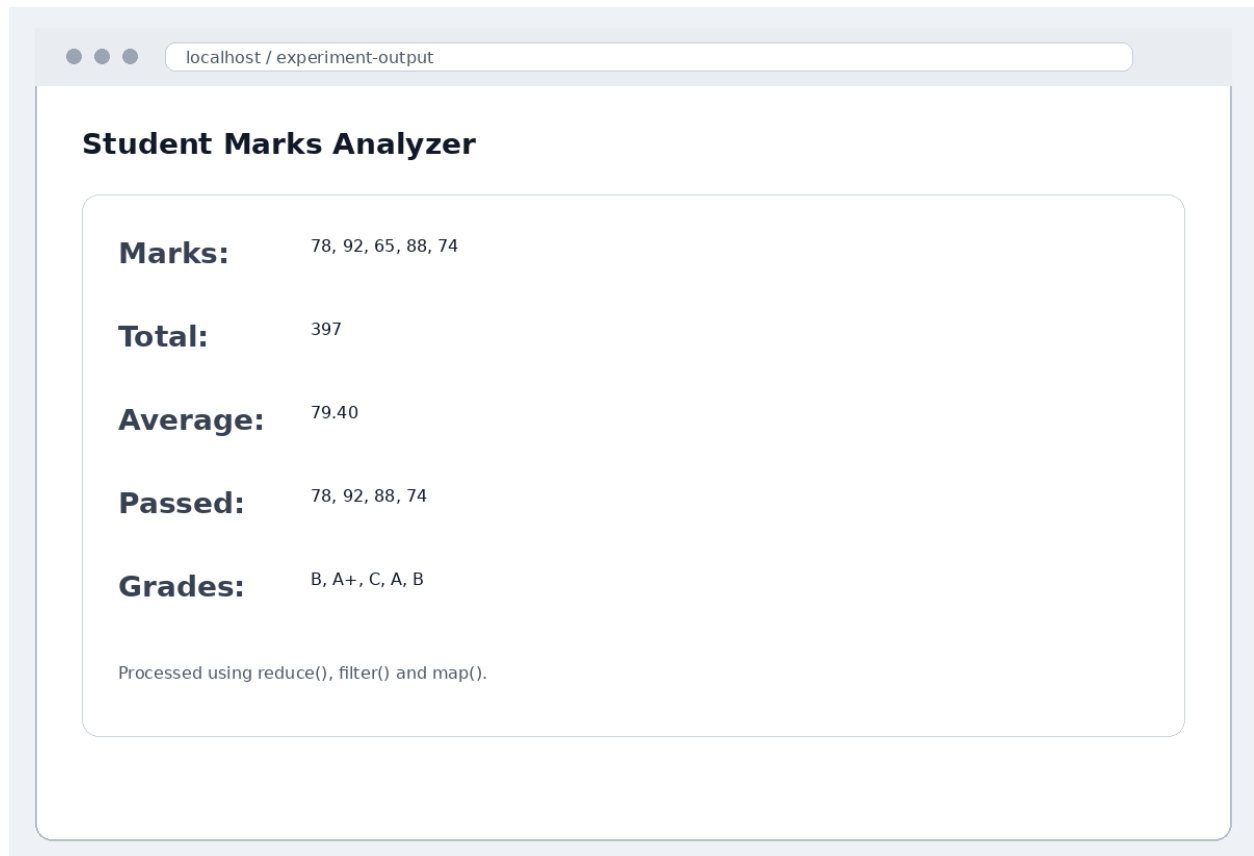
The screenshot shows a web browser window with the title 'localhost / experiment-output'. The page content is titled 'Student Marks Analyzer'. It displays the following information:

Marks:	69, 70, 79, 89, 91
Total:	398
Average:	79.60
Passed:	70, 79, 89, 91
Grades:	C, B, B, A, A+

At the bottom, it states: 'Processed using reduce(), filter() and map()'.

8. OUTPUT

Browser Output (Screenshot):



9. RESULT

Thus, JavaScript arrays and functions were successfully implemented. The program calculates total and average marks, filters passed marks and generates grades using `reduce()`, `filter()` and `map()`.